Certified Coverage of Exponential Search Spaces

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Idea. Consider a search problem whose naive state space has size 2^n. Some states may admit a compact structural representation; others may be proposed for elimination by a learned model and certified exactly; still others may be represented by exact summaries of paired groups. Only the residual must be searched explicitly. The central quantity is the *growth of what remains uncovered*, together with the cost of obtaining coverage.

A coverage model

Let $n \in \mathbb{N}$ be the problem size and $\mathcal{S}_n$ its set of candidate states, with

$$|\mathcal{S}_n| = 2^n.$$

A state is *covered* when an exact representation or a sound exclusion removes the need to search it individually. We use three mechanisms on disjoint parts of the space; all state counts below count distinct states.

First, a structural procedure identifies a set $K_n \subseteq \mathcal{S}_n$ of size

$$k(n) = |K_n|$$

that admits a compact representation. Assume $1 \le k(n) \le 2^n$, so the logarithm below is defined. In the simplest logarithmic model, those $k(n)$ states are replaced by a description of cost

$$C_{\text{comp}}(n) = \log_2 k(n).$$

This is deliberately an abstract assumption: it describes the cost once such a representation is available, not a universal property of arbitrary subsets.

Second, a learned classifier proposes regions outside K_n for removal. Let $\delta(n)$ count the distinct proposed states, counting overlaps only once. The classifier cannot delete states directly: a sound verifier certifies only regions whose omission leaves the search answer unchanged. Let $a(n) \in [0, 1]$ be the fraction of those $\delta(n)$ states certified by the verifier, with $a(n) = 0$ if $\delta(n) = 0$. The safely eliminated count is

$$\delta_c(n) = a(n)\,\delta(n).$$

Prediction error therefore affects efficiency, not correctness: uncertified proposals simply remain in the residual problem.

Third, pair groups of remaining states and replace each pair by an exact summary for the search question. Let $p(n)$ count the distinct states represented by these paired summaries, excluding states already counted in $k(n)$ or $a(n)\delta(n)$. Let $Q_{\text{pair}}(n)$ be the cost of finding pairs, constructing and certifying their summaries, and using the resulting representation.

The explicit residual is

$$r(n) = 2^n - k(n) - a(n)\delta(n) - p(n). \tag{1}$$

Let $P(n)$ collect structural discovery and learned proposal and verification work, excluding the pairing cost $Q_{\text{pair}}(n)$ and the structural description charge $\log_2 k(n)$. All costs use the same units. Normalizing explicit search to one unit per residual state gives the modeled runtime

$$T(n) = P(n) + Q_{\text{pair}}(n) + r(n) + \log_2 k(n). \tag{2}$$

Thus

$$\begin{aligned} T(n) = P(n) + Q_{\text{pair}}(n) + 2^n - k(n) \\ - a(n)\delta(n) - p(n) + \log_2 k(n). \end{aligned} \tag{3}$$

The separation is important. Large covered regions help only if discovering, representing, and certifying them is cheap.

The certified coverage theorem

Define the total *certified coverage*

$$C(n) = k(n) + a(n)\delta(n) + p(n),$$

so that $r(n) = 2^n - C(n)$. Here and below, $f(n) = \text{poly}(n)$ means $f(n) = O(n^d)$ for some fixed $d \geq 0$.

Theorem 1 (Certified coverage criterion). Assume $P(n) + Q_{\text{pair}}(n) = \text{poly}(n)$, $1 \leq k(n) \leq 2^n$, and the residual is searched in time proportional to $r(n)$. Then

$$T(n) = \text{poly}(n) \quad \Longleftrightarrow \quad 2^n - C(n) = \text{poly}(n).$$

Equivalently, polynomial modeled runtime occurs exactly when

$$C(n) = 2^n - \text{poly}(n).$$

Proof. From (2),

$$T(n) = P(n) + Q_{\text{pair}}(n) + \big(2^n - C(n)\big) + \log_2 k(n).$$

Because $k(n) \leq 2^n$,

$$0 \leq \log_2 k(n) \leq n.$$

The overhead and logarithmic terms are polynomially bounded. Hence $T(n)$ is polynomial exactly when the residual $2^n - C(n)$ is polynomially bounded. □

The theorem gives a sharp asymptotic warning: eliminating a very large *percentage* of an exponential space need not change its asymptotic order.

Corollary 1 (Fixed uncovered fraction). If there exists a constant $\varepsilon > 0$ such that, for infinitely many n,

$$C(n) \leq (1 - \varepsilon)2^n,$$

then, for those n,

$$T(n) \geq r(n) \geq \varepsilon 2^n.$$

Thus 90%, 99%, or even 99.999% coverage is asymptotically insufficient if the uncovered fraction stays bounded away from zero. Polynomial behavior requires

$$\frac{C(n)}{2^n} = 1 - \frac{\mathrm{poly}(n)}{2^n},$$

so the uncovered fraction must vanish exponentially fast, up to polynomial factors.

Differential form

For intuition, choose differentiable extensions of k, δ, a, p, P, and Q_{pair} from integer to real n, with $k > 0$, and define r, T there by (1)–(2). Primes denote derivatives with respect to n; the equations below describe this chosen interpolation. Differentiating (1) gives

$$\frac{dr}{dn} = (\ln 2)2^n - k'(n) - a(n)\delta'(n) - a'(n)\delta(n) - p'(n). \tag{4}$$

The terms are raw state creation, structural coverage, new certified proposals, improving certification rate, and paired coverage, respectively.

In the next two displays, unprimed k, a, δ abbreviate their values at n. Differentiating the full cost gives

$$\begin{aligned}\frac{dT}{dn} = (\ln 2)2^n - \left(1 - \frac{1}{k \ln 2}\right) k' - a\delta' - a'\delta - p' \\ + P'(n) + Q'_{\mathrm{pair}}(n).\end{aligned} \tag{5}$$

When the correction $k'/(k \ln 2)$ is small relative to the terms of interest, this gives the approximate balance law

$$\begin{aligned}T'(n) \approx (\ln 2)2^n - k' - a\delta' - a'\delta - p' \\ + P'(n) + Q'_{\mathrm{pair}}(n).\end{aligned} \tag{6}$$

Equivalently, since $r(n) = 2^n - C(n)$,

$$r'(n) = (\ln 2)2^n - C'(n).$$

Integrating from a reference scale n_0,

$$r(n) = r(n_0) + \int_{n_0}^{n} \left[(\ln 2)2^s - C'(s)\right] ds. \tag{7}$$

This form makes the central picture transparent: at integer n, polynomial behavior requires the cumulative gap between exponential state creation and certified coverage growth to remain polynomial in n.

Why certification matters

Suppose a naive learned-pruning model deletes $\delta(n)$ states and $e(n) \in [0, 1]$ is the fraction deleted incorrectly. Exact recovery would require revisiting $e(n)\delta(n)$ states. When $\delta(n) = \Theta(2^n)$, any fixed positive error fraction leaves exponential recovery work. Keeping that work polynomial requires, for some fixed $d \geq 0$,

$$e(n)\delta(n) = O(n^d), \qquad \text{hence} \qquad e(n) = O\left(\frac{n^d}{2^n}\right).$$

Thus ordinary constant-error prediction is not enough for exact bulk deletion of an exponential region.

Certification avoids that failure mode. The learned model becomes a *proposal mechanism*: it suggests large regions that may be irrelevant, while an exact verifier alone has authority to delete them. A poor classifier can waste computation, but it cannot make the final algorithm incorrect. In this formulation, learning changes δ and the acceptance function a; correctness is carried by verification.

The cost of pairing

For disjoint paired groups $B_0, B_1 \subseteq \mathcal{S}_n$, let $s(B)$ be a summary of group B, and let Merge combine two summaries. Exact pairing requires

$$s(B_0 \cup B_1) = \mathrm{Merge}\big(s(B_0), s(B_1)\big),$$

Here *exact* means that replacing a paired group by its summary leaves the final answer unchanged when combined with all other covered groups and the residual. A complete explicit pairing tree over 2^n states has $2^n - 1$ merges, even though only one group remains. Thus pairing helps asymptotically only when a compact rule or reuse of summaries keeps $Q_{\mathrm{pair}}(n)$ polynomial. Counting covered states without this cost would hide the original exponential work.

Scope and the remaining difficulty

The coverage criterion is a theorem about the model; it does not assert that every concrete search problem admits such coverage. The hard question is whether, for a given problem, one can simultaneously achieve

$$P(n) + Q_{\mathrm{pair}}(n) = \mathrm{poly}(n) \qquad \text{and} \qquad 2^n - C(n) = \mathrm{poly}(n).$$

Cardinality alone cannot answer this: exponentially many states can sometimes be described by a short rule. The substantive lower- or upper-bound problem is therefore about the *cost of discovering and certifying structure*, not merely the number of states represented by that structure.

Two companion notes follow this question through subset sum: first as a stress test of exact pairing and learned proposals, then as a probability-valued reachable-target problem. The latter changes the objective and uses its own certified bounds.

General representation cost

The logarithmic term is the motivating special case, not an essential part of the coverage theorem. Let $\rho(n, k)$ be the cost to store and use a structural representation covering k states at problem size n. Then write

$$T(n) = P(n) + Q_{\mathrm{pair}}(n) + r(n) + \rho(n, k(n)).$$

Whenever $P(n) + Q_{\mathrm{pair}}(n)$ and $\rho(n, k(n))$ are polynomially bounded, the same proof gives

$$T(n) = \mathrm{poly}(n) \quad \Longleftrightarrow \quad r(n) = 2^n - C(n) = \mathrm{poly}(n).$$

Thus the essential object is the uncovered residual; $\log_2 k$ is simply the strongest compression regime considered here. This also separates two questions that should not be conflated: how many states a representation covers, and how costly that representation is to discover and manipulate.

Summary. Exponential search becomes a competition between state-space growth and certified coverage. Structural compression and exact pairing represent states collectively; learning proposes further regions; certification makes eliminations exact; and the residual determines explicit work. The governing condition is simple:

$$\text{polynomial overhead} + \text{polynomial residual} \implies \text{polynomial modeled runtime}.$$

The differential form records the same statement dynamically: coverage must track exponential state creation closely enough that their accumulated gap is only polynomial.